

LiDAR Camera Calibration User Guide

What This Application Does

This application calibrates a camera and a LiDAR sensor mounted on the same robot or vehicle — that is, it works out the precise position and orientation of the camera relative to the LiDAR. This calibration is required before the two sensors' data can be combined (for example, to color a LiDAR point cloud using the camera image, or to project LiDAR points onto the camera view).

You do not need to write any code or use a terminal. Everything is done through a single window with four buttons, one for each step of the calibration process.

The only thing you need to provide is a short recording (“bag”) of the camera and LiDAR looking at the same scene at the same time.

Getting Started

Requirements

- A 64-bit Linux computer
- Your recorded data (see “Preparing Your Data” below)

Running the Application

1. Locate the application file — it will have a name like `direct-lidar-camera-calibration-<jazzy/bionic/other version>-x86_64.AppImage`.
2. Make sure it's marked as runnable. If you're using a file manager, right-click the file → **Properties** → **Permissions** → check “**Allow executing file as program.**” If you're using a terminal, run the command below.
3. Double-click the file (or run it from a terminal).

```
chmod +x direct-lidar-camera-calibration-jazzy-x86_64.AppImage
```

The application window opens directly — there's nothing to install.

Preparing Your Data

The application reads **rosbag2** recordings — the standard recording format used in robotics. Each recording (each “bag”) should contain, at minimum:

- A camera image topic
- A LiDAR point cloud topic

- A camera info topic

You can provide **one or more** bag recordings. More recordings from different viewpoints generally produce a more accurate calibration.

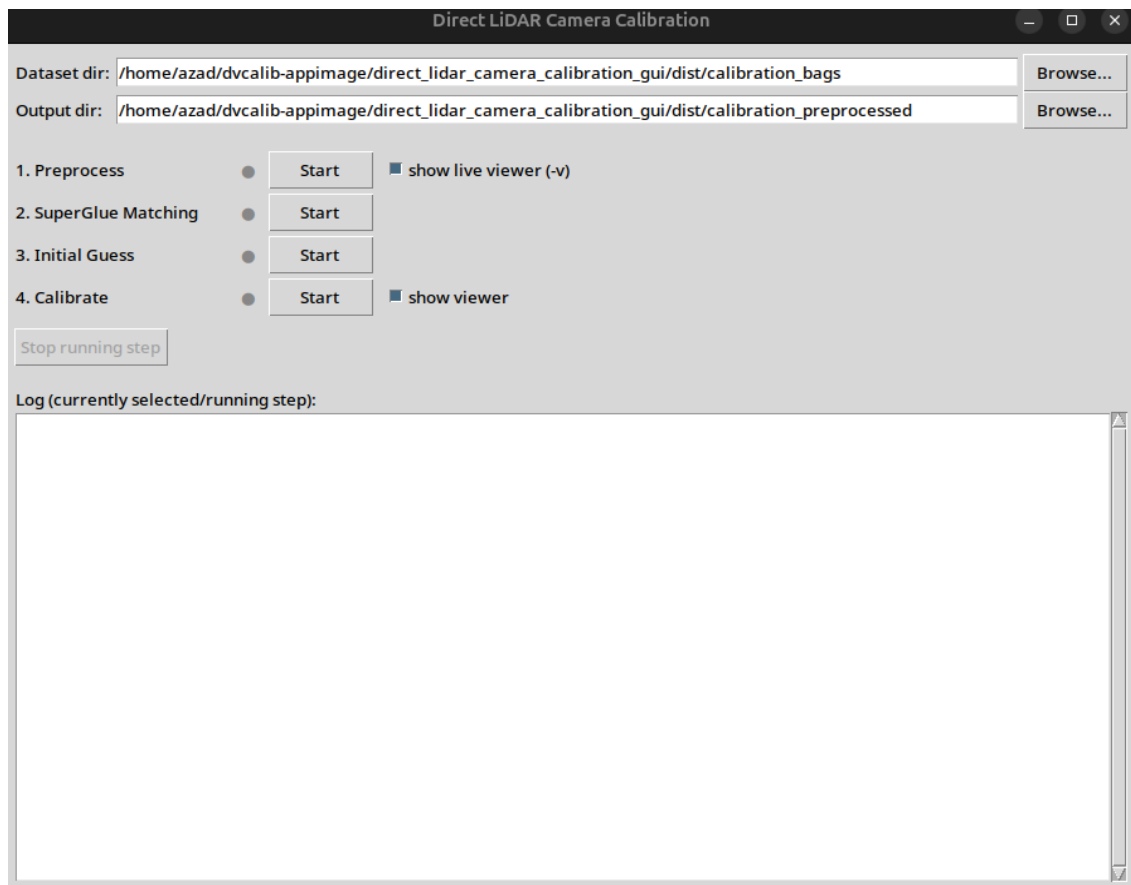
Put all your bag folders inside a single folder — for example:

```
my_calibration_data/  
├── bag1/  
├── bag2/  
└── bag3/
```

You'll point the application at the `my_calibration_data` folder (the one containing the bag folders), not at an individual bag.

Tip: If a folder named `calibration_bags` sits right next to the application file, it will be picked automatically as the default Dataset directory — you can just drop your bags in there and skip typing a path.

The Main Window



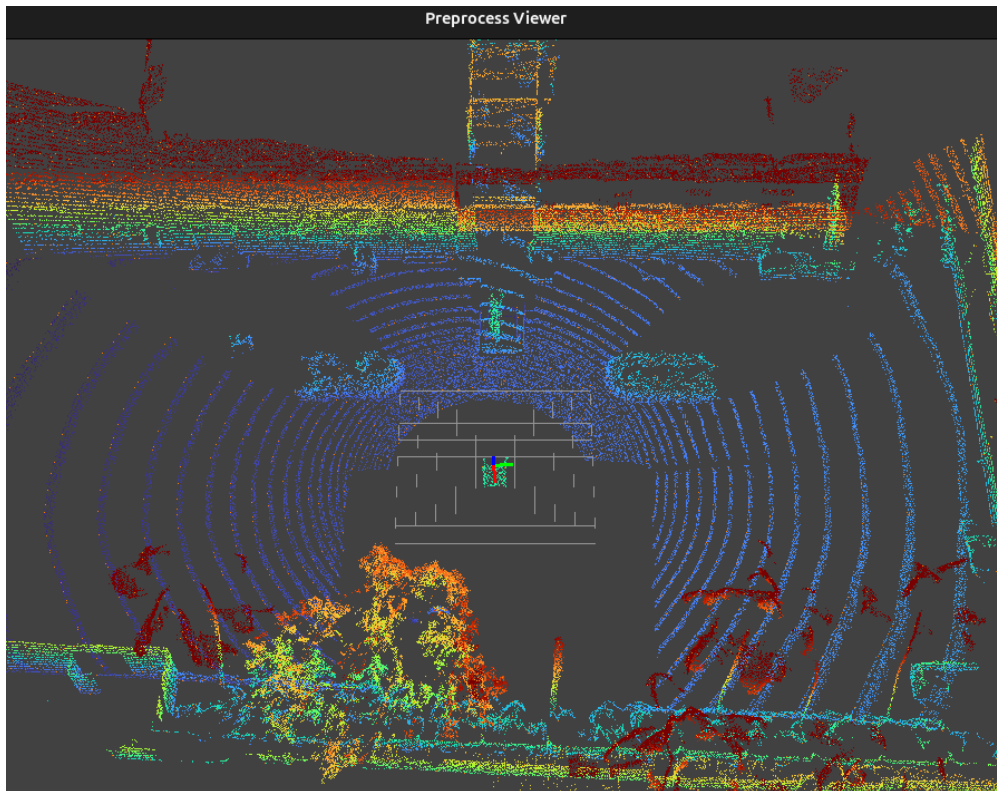
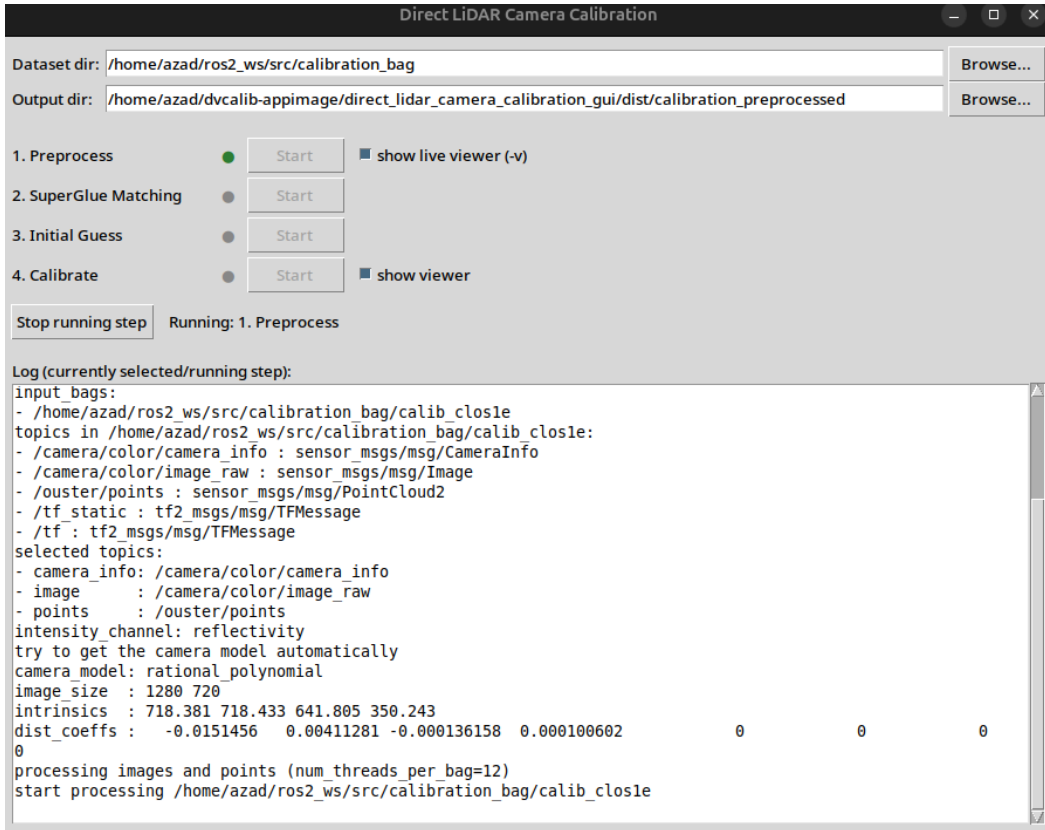
Area	What it's for
Dataset dir	The folder containing your bag recordings (see above). Use Browse... to pick it, or type the path directly.
Output dir	Where the application writes its results. It's created automatically if it doesn't exist — you don't need to create it yourself.
1. Preprocess / 2. SuperGlue Matching / 3. Initial Guess / 4. Calibrate	The four steps of the calibration, explained one by one below. Each has its own Start button and a status dot next to it.
Status dot	Shows the state of that step: grey = not started yet, green = running now (or finished successfully), red = failed, or was stopped.
show live viewer (-v) / show viewer	Checkboxes next to steps 1 and 4. When checked, a separate window pops up showing a live 3D view while that step runs (see “About the Viewer Windows”).
Stop running step	Cancels whichever step is currently running.
Log	Shows the live text output of whichever step you most recently started — useful for seeing progress, or for understanding what went wrong if a step fails.

Step-by-Step: Running a Calibration

- Set the Dataset directory to the folder containing your bag recordings, and the Output directory to wherever you want the results saved (or just leave the defaults).
- Click “Start” next to “1. Preprocess.”

This step reads your bag recordings and extracts images and dense point clouds from them.

- If **show live viewer** is checked, a separate window will pop up showing the point cloud being built up live. This is normal — leave it be, it will close on its own when the step finishes.
- This is typically the slowest step — depending on how much data you recorded, it can take anywhere from under a minute to several minutes. Watch the status dot: it turns green once this step succeeds.



6. Click “Start” next to “2. SuperGlue Matching.”

This step finds matching features between the camera image and the LiDAR data. Usually quick.

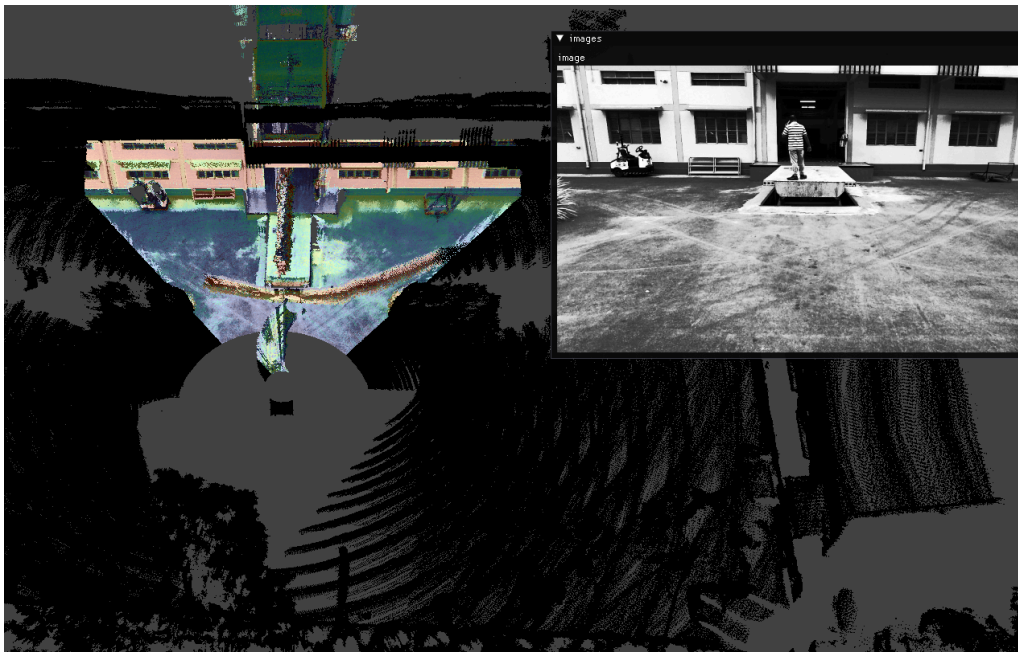
7. Click “Start” next to “3. Initial Guess.”

This step computes a rough first estimate of the camera-to-LiDAR alignment from those matches. Usually quick.

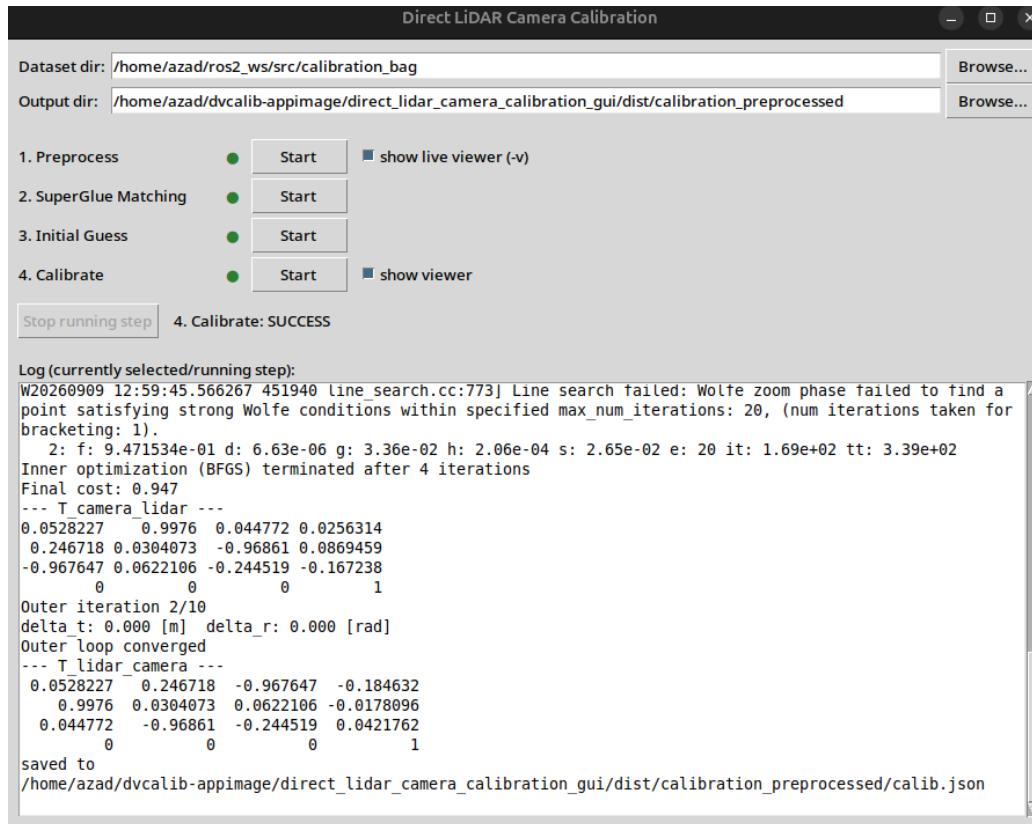
8. Click “Start” next to “4. Calibrate.”

This is the fine-tuning step — it refines the rough estimate into the final, accurate calibration. It's usually the second-slowest step (can take a few minutes).

- If **show viewer** is checked, a window pops up showing the LiDAR points overlaid on the camera image, updating live as the calibration improves. This is the best way to visually confirm the result looks correct — the LiDAR points should line up with edges and objects visible in the camera image.



9. Wait for the status dot next to “4. Calibrate” to turn green. The calibration is now complete.



Where to Find Your Result

Once step 4 finishes successfully, open your Output directory and look for a file named `calib.json`. This file contains the final calibration result, under a field named `T_lidar_camera` — the transform describing the camera's position and orientation relative to the LiDAR. This is the file you hand off to whatever downstream system needs the calibration (you won't normally need to open or edit it yourself).

If Something Goes Wrong

- **A status dot turns red:** the step failed. Look at the Log area — it shows the live output from that step, including any error message. Each step also writes a full log file into `<your output directory>/logs/`, in case you want to look more closely or send it to support.
- **Common causes of failure:**
 - The Dataset directory doesn't actually contain any valid bag recordings.
 - You started a later step (e.g. “Calibrate”) before an earlier one (e.g. “Preprocess”) had actually finished successfully.
- **A step seems stuck:** some steps can genuinely take several minutes — check the Log area for signs of ongoing activity before assuming it's frozen. If you do want to cancel it, click **Stop running step**.

